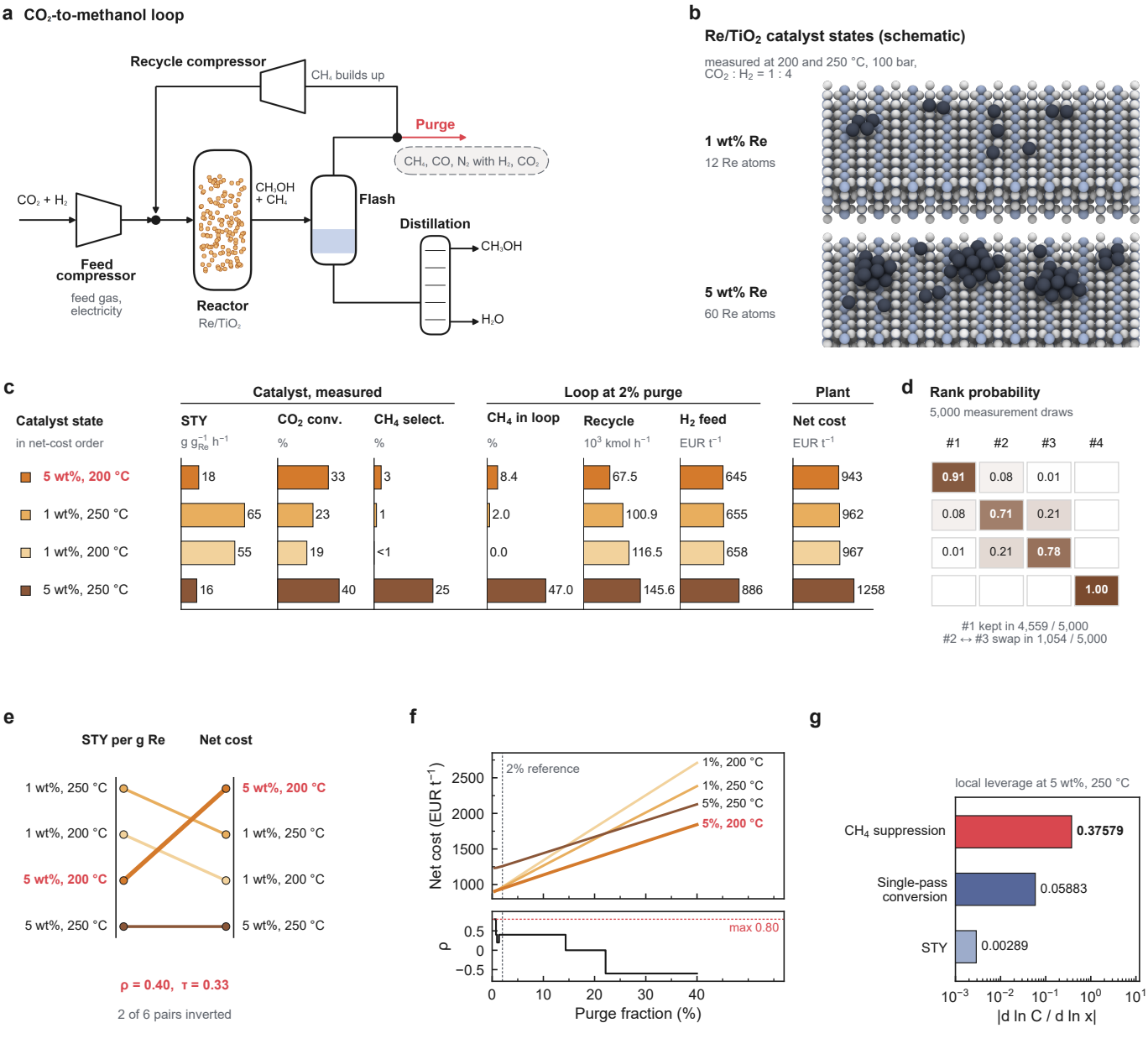

a CO₂-to-methanol loop
Recycle compressor
CH₄ builds up
Purge
CH₄, CO, N₂ with H₂, CO₂
CO₂ + H₂
CH₃OH + CH₄
Flash
Distillation
CH₃OH
H₂O
Feed compressor
feed gas, electricity
Reactor
Re/TiO₂
b
Re/TiO₂ catalyst states (schematic)
measured at 200 and 250 °C, 100 bar, CO₂ : H₂ = 1 : 4
1 wt% Re
12 Re atoms
5 wt% Re
60 Re atoms
c
Catalyst, measured
Loop at 2% purge
Plant
Catalyst state
in net-cost order
STY
g g_Re^-1 h^-1
CO₂ conv.
%
CH₄ select.
%
CH₄ in loop
%
Recycle
10³ kmol h⁻¹
H₂ feed
EUR t⁻¹
Net cost
EUR t⁻¹
5 wt%, 200 °C 18 33 3 8.4 67.5 645 943
1 wt%, 250 °C 65 23 1 2.0 100.9 655 962
1 wt%, 200 °C 55 19 <1 0.0 116.5 658 967
5 wt%, 250 °C 16 40 25 47.0 145.6 886 1258
d
Rank probability
5,000 measurement draws
#1 #2 #3 #4
0.91 0.08 0.01
0.08 0.71 0.21
0.01 0.21 0.78
1.00
#1 kept in 4,559 / 5,000
#2 ↔ #3 swap in 1,054 / 5,000
e
STY per g Re
Net cost
1 wt%, 250 °C
1 wt%, 200 °C
5 wt%, 200 °C
5 wt%, 250 °C
5 wt%, 200 °C
1 wt%, 250 °C
1 wt%, 200 °C
5 wt%, 250 °C
ρ = 0.40, τ = 0.33
2 of 6 pairs inverted
f
Net cost (EUR t⁻¹)
2% reference
1%, 200 °C
1%, 250 °C
5%, 250 °C
5%, 200 °C
ρ
max 0.80
Purge fraction (%)
g
local leverage at 5 wt%, 250 °C
CH₄ suppression
0.37579
Single-pass conversion
0.05883
STY
0.00289
|d ln C / d ln x|